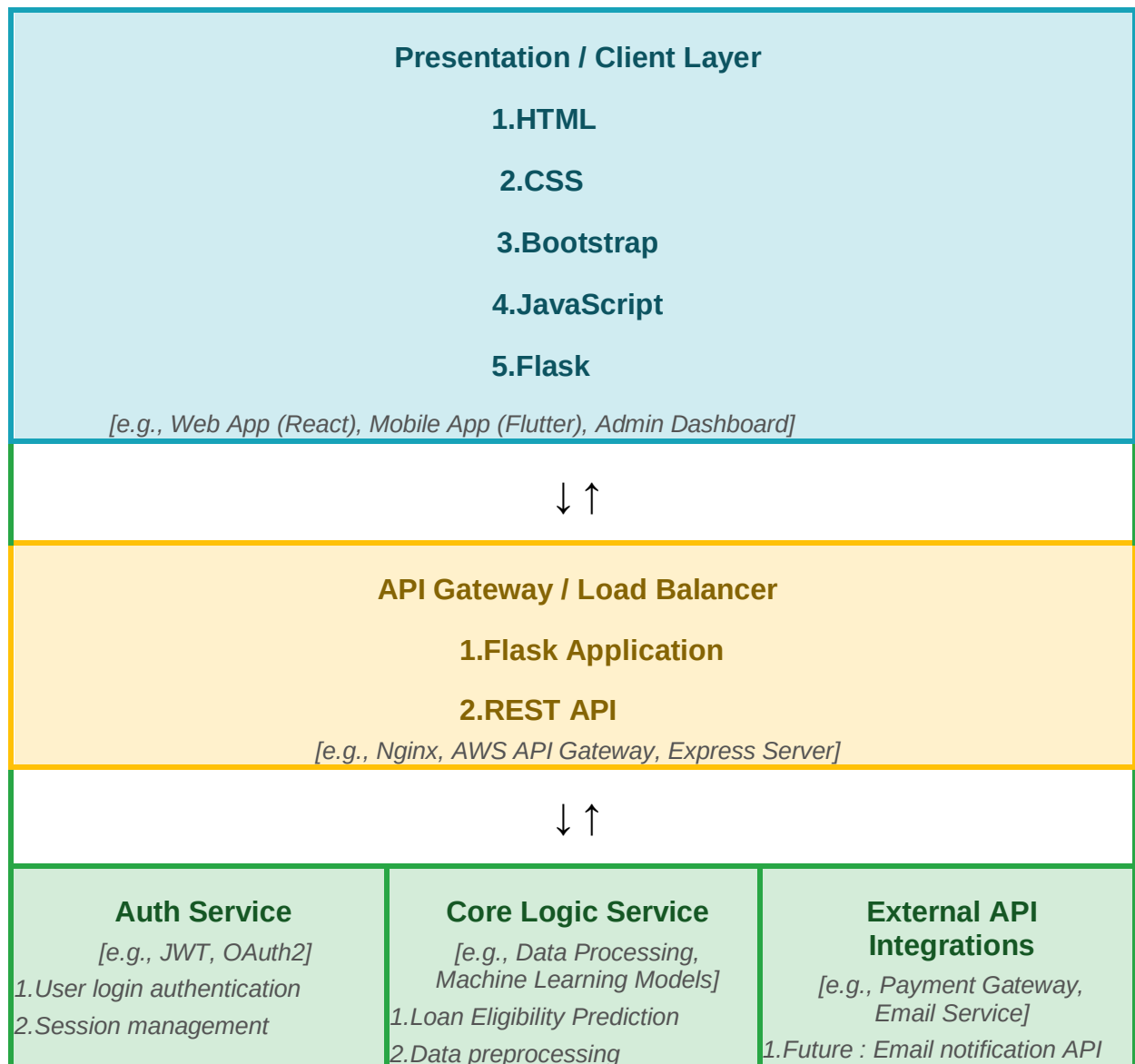


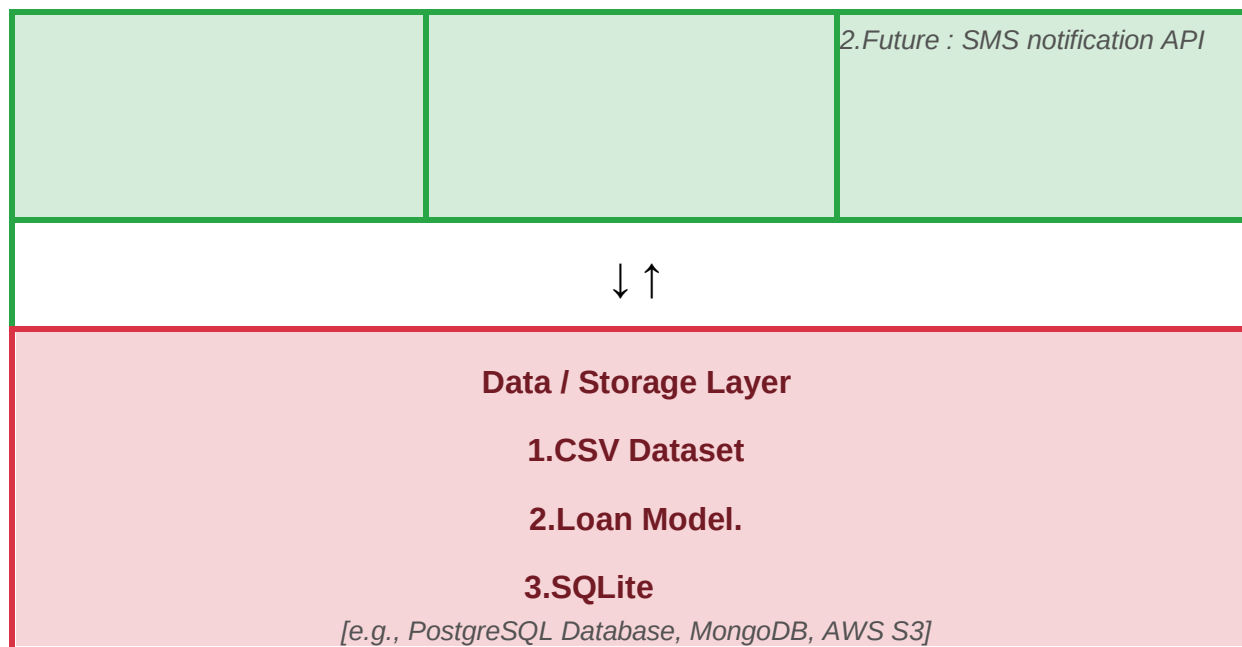
## Solution Architecture

|                      |                                                   |
|----------------------|---------------------------------------------------|
| <b>Date</b>          | 06 July 2026                                      |
| <b>Team ID</b>       |                                                   |
| <b>Project Name</b>  | Smart Lender - Loan Eligibility Prediction System |
| <b>Maximum Marks</b> | 5 Marks                                           |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





## Component Description Table

| Component Name              | Description / Role in Architecture                                | Technologies Used                               |
|-----------------------------|-------------------------------------------------------------------|-------------------------------------------------|
| <i>[Presentation Layer]</i> | <i>Collects user loan details and displays prediction results</i> | <i>HTML,CSS, Bootstrap, javascript, Flask</i>   |
| <i>[API Gateway]</i>        | <i>Receives user requests and forwards them to the ML model</i>   | <i>Flask ,REST API</i>                          |
| <i>[Microservices]</i>      | <i>Performs data preprocessing</i>                                | <i>Python,Pandas Numpy,Scikitlearn, XGBoost</i> |
| <i>[Database]</i>           | <i>Stores user details, datasets and trained model</i>            | <i>SQLite and MySQL,CSV, Joblib</i>             |